

Forecast Evaluation with Observational Uncertainty: A Cramér-Distance Perspective

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Abstract

Armed-conflict fatality counts are uncertain. UCDP data—the standard source for conflict forecasting—systematically underreport deaths, with estimated underreporting exceeding 100% for low-fatality events—meaning the true count may be more than twice the recorded figure (Vesco et al., 2026). As the field moves from point-estimate predictions toward full predictive distributions (Hegre et al., 2025), the question of how to evaluate these distributions becomes critical. The dominant evaluation metric—the Continuous Ranked Probability Score (CRPS)—assumes the observed fatality count is exact. This creates a bias in model selection: CRPS rewards forecasts that concentrate on the reported number over forecasts that honestly represent the uncertainty around it. I show that replacing this assumption with a distribution reflecting what is actually known about the true value yields a well-established metric—the Cramér distance, which is simply the standard CRPS generalised from a single observed value to an uncertain observation—that reduces to standard CRPS when observations are precise. I provide three practical recipes for building the observation distribution from sources available to conflict researchers: from UCDP quality bounds, from the empirically grounded uncertainty model of Vesco et al. (2026), and from a calibrated parametric default. On 19,397 UCDP cell-months, the reformulation reverses the model ranking in 39% of cases; on operational predictions from HydraNet (von der Maase, 2025), a spatio-temporal deep-learning model for conflict forecasting, 99.4% of cases where the standard metric favours overconfidence are reversed. The metric’s reliability depends on the quality of the observation distribution; I characterise the conditions under which mis-specification introduces bias and verify consistency numerically across three distributional families.

Significance. Forecast competitions and model selection procedures that use CRPS as their ranking metric—such as the 2023/24 VIEWS prediction challenge (Hegre et al., 2025)—implicitly assume observations are exact. When this assumption is violated, competition rankings can systematically favour overconfident models. I show that accounting for observational uncertainty

in forecast evaluation changes which forecast version is judged best, and demonstrate this on both synthetic and operational conflict forecasts (§5). The argument applies to any domain where observations carry non-trivial uncertainty (Bessac and Naveau, 2021).

1 Introduction

Armed-conflict fatality counts are derived from media reports, witness accounts, and analyst judgement; the reported value is a noisy estimate of an underlying truth that no single source observes exactly. UCDP—the standard source for conflict forecasting—explicitly provides low/best/high uncertainty bounds per event (Uppsala Conflict Data Program, 2025), and estimated underreporting exceeds 100% for low-fatality events (Vesco et al., 2026). As the field moves toward full predictive distributions—the 2023/24 VIEWS prediction challenge required submissions as distributional forecasts rather than point estimates (Hegre et al., 2025)—the question of how to evaluate these distributions against uncertain observations becomes critical.

The dominant evaluation metric is the Continuous Ranked Probability Score (CRPS; Matheson and Winkler, 1976; Hersbach, 2000). It measures how well a forecast distribution matches an observation: the closer the forecast’s cumulative distribution function (CDF) F_{pred} is to the observation, the lower the score:

$$\text{CRPS}(F_{\text{pred}}, y) = \int_{-\infty}^{\infty} (F_{\text{pred}}(z) - \mathbf{1}\{y \leq z\})^2 dz. \quad (1)$$

CRPS is strictly proper (Gneiting and Raftery, 2007)—a forecaster who reports the distribution she actually believes always scores at least as well as one who misreports, so the metric cannot be gamed. This property has made CRPS central to forecast verification in meteorology (Candille and Talagrand, 2005; Jordan et al., 2019), hydrology (Laio and Tamea, 2007), economics (Gneiting and Ranjan, 2011), and conflict forecasting (Hegre et al., 2019).

The step indicator $\mathbf{1}\{y \leq z\}$ encodes an implicit assumption: that the observation y is the true value, known without error. When this assumption is violated, CRPS rewards forecast distributions that concentrate probability mass near the reported value—even when a wider distribution would better represent what is actually known about the underlying truth. Two versions of the same forecast can receive opposite rankings depending on whether they calibrate to the reported point (rewarded by CRPS) or to the plausible range of underlying values (penalised by CRPS). This is not a flaw in the propriety theorem—the theorem concerns draws from the true distribution—but a practical consequence of using a noisy proxy for the truth as if it were exact.

The problem of scoring forecasts against uncertain observations has been addressed from several principled angles, primarily in meteorology where rain gauges and satellite retrievals carry known measurement error: conditional-expectation corrections (Bröcker and Smith, 2007), Bayesian error models (Ferro, 2017b), distributional CRPS analysis (Bessac and Naveau, 2021; Bolin and Wallin, 2023), and information-theoretic frameworks (Weijs et al., 2011). Separately, Thorarinsdottir et al. (2013) define the Integrated Quadratic Distance (IQD)—the L^2 distance

between two CDFs, i.e. the Cramér distance (Cramér, 1946)—as a proper divergence for comparing non-degenerate distributions; classical CRPS is the special case where one CDF degenerates to a step function. I compare these approaches to the present reformulation in §6.

I am not proposing a new metric—the Cramér distance has been established since at least the mid-twentieth century (Cramér, 1946) and its use as a verification divergence is known (Thorarinsdottir et al., 2013; Székely and Rizzo, 2013). What I contribute is an operational examination of this distance *from the perspective of conflict forecasting*, where observational uncertainty is the norm rather than the exception, and where the practical question—when does acknowledging that uncertainty change which model we would select?—has not been investigated.

My contribution is threefold, all operational rather than theoretical:

1. Three *constructors*—operational recipes for building F_{obs} from available data—tailored to the data sources common in conflict and social-science forecasting: from explicit dataset quality bounds, from the empirically estimated Reported-Value Inflated (RVI) Gumbel mixture of Vesco et al. (2026), and from a parametric default (§3).
2. A characterisation of the parameter regime in which replacing the step indicator with a smooth F_{obs} produces a different ranking of forecast versions from classical CRPS, and the regime in which it does not—demonstrated on controlled examples, UCDP conflict data, and operational HydraNet predictions (von der Maase, 2025) (§5).
3. Numerical verification of consistency under stochastic observations, with an empirical characterisation of the conditions under which mis-specification of F_{obs} introduces bias (§4, §6).

The reformulation reduces to classical CRPS when observational uncertainty is zero, so it generalises rather than replaces the standard metric.

2 Classical CRPS and Its Implicit Assumption

The indicator $\mathbf{1}\{y \leq z\}$ in Eq. 1 jumps from 0 to 1 at $z = y$. The integrand is the squared distance between the forecast CDF and a hard step function at the observation.

Strict propriety. A scoring rule $S(F, y)$ is strictly proper if, for any F and any true distribution G ,

$$\mathbb{E}_{y \sim G}[S(F, y)] \geq \mathbb{E}_{y \sim G}[S(G, y)], \quad (2)$$

with equality if and only if $F = G$ almost surely. Gneiting and Raftery (2007) prove that CRPS satisfies this property: a forecaster who reports the true distribution scores best in expectation.

The implicit assumption. The propriety theorem treats y as a draw from G . It says nothing about the relationship between y and the underlying physical or social quantity that G describes. The CRPS formula treats the reported y *as if* it were the true value: the indicator function has no width.

For some data sources this is correct. A coin flip outcome is exact. A digital weather station’s temperature has measurement uncertainty small relative to forecast uncertainty, so treating the reading as exact is harmless. But for other data sources it is not. Conflict fatality counts come from coding pipelines that aggregate media reports, witness accounts, and expert judgement. The UCDP Georeferenced Event Dataset (Uppsala Conflict Data Program, 2025) explicitly provides low/best/high estimates per event and documents systematic underreporting in remote regions. The reported best estimate is a noisy summary of an underlying truth—and similar observation uncertainty arises in disease surveillance (Gibbons et al., 2014) and other domains where data are compiled from heterogeneous sources.

Why the assumption matters. When the assumption is violated, CRPS systematically rewards narrow, overconfident forecast distributions over wider, more honest ones that better represent the underlying uncertainty. Concretely: if UCDP records 12 deaths in a cell-month but the true count is plausibly between 8 and 30, a forecast that says “exactly 12” scores better under classical CRPS than one that spreads probability across that range—even though the wider forecast better represents what is actually known. I quantify this bias in §5.

The existing literature includes one well-known modification in this direction: threshold- and quantile-weighted CRPS (Gneiting and Ranjan, 2011), which rescales the integrand to emphasise specific regions of the distribution. This addresses a different problem—the Forecaster’s Dilemma (Lerch et al., 2017), in which conditioning forecast evaluation on extreme observations systematically discredits skillful forecasts. In conflict forecasting, this means that evaluating models only on high-fatality months rewards models that predict high fatalities uniformly; weighted scoring rules are a complementary, not competing, tool. Crucially, weighted CRPS does not lift the assumption that the observation is known precisely. The reformulation I propose is orthogonal: it modifies the indicator function rather than the weight on the integrand.

Visual illustration. Figure 1 illustrates the CDF comparison underlying classical CRPS (left) and the reformulated metric (right) for the same forecast and the same observed value. In the classical version, F_{obs} is a step at y , and the area between F_{pred} and F_{obs} is large unless F_{pred} is very narrow. In the reformulated version, F_{obs} is a smooth CDF reflecting observational uncertainty, and the area is smaller for the same forecast, because the forecast is no longer asked to match an impossibly precise step.

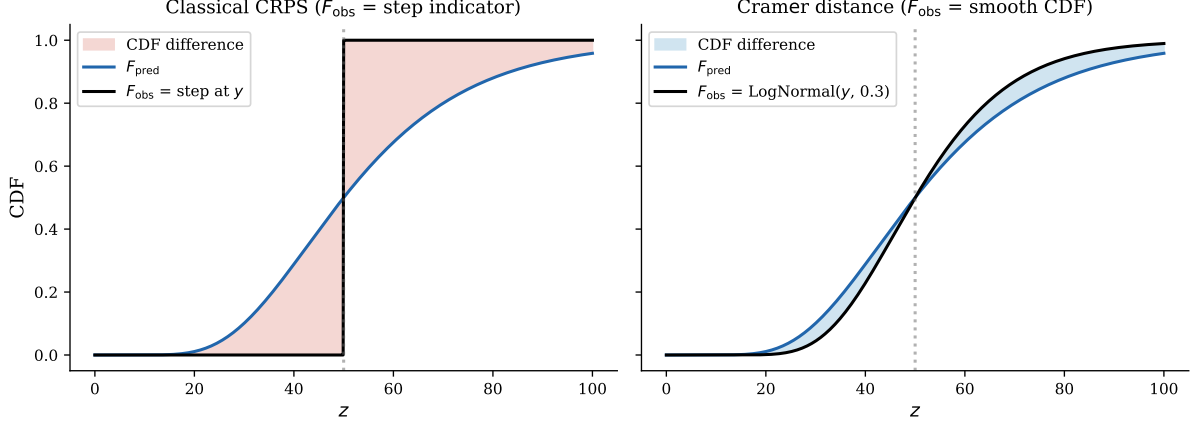


Figure 1: CDF comparison under two definitions of F_{obs} . Left: classical CRPS uses a step indicator; the shaded area between the smooth forecast and the hard boundary is large. Right: the Cramér-distance reformulation uses a smooth F_{obs} reflecting observational uncertainty; a moderately wide forecast is no longer penalised relative to an overconfident one. The shaded region shows the gap $F_{\text{pred}}(z) - F_{\text{obs}}(z)$; the Cramér-distance integrand is its square.

3 The Cramér-Distance Reformulation

Reformulation. The fix is straightforward: instead of comparing the forecast against a hard step at the reported value, I compare it against a smooth distribution that represents what is actually known about the observation. Concretely, I replace the step indicator in the CRPS integrand with a smooth observational CDF $F_{\text{obs}}(z)$:

$$\text{CRPS}_{\text{obs}}(F_{\text{pred}}, F_{\text{obs}}) = \int_{-\infty}^{\infty} (F_{\text{pred}}(z) - F_{\text{obs}}(z))^2 dz. \quad (3)$$

When F_{obs} degenerates to a step function at the reported value y , the integral reduces exactly to classical CRPS. When F_{obs} has positive variance, the integrand is smaller for forecasts that match F_{obs} and larger for forecasts that deviate in either direction.

Identification with the Cramér distance. The integral in Eq. 3 is a known mathematical object—the Cramér distance between F_{pred} and F_{obs} , measuring the total squared area between two probability curves (Cramér, 1946).¹ In forecast verification, Thorarinsdottir et al. (2013) define the same integral as the *Integrated Quadratic Distance* (IQD) and apply it to evaluate climate model output against reanalysis distributions—both non-degenerate CDFs. Classical CRPS is the special case where F_{obs} degenerates to a step function at a single point; Eq. 3 is the general form applied to a non-degenerate F_{obs} .

Constructing F_{obs} . The proposal is only as useful as the choice of F_{obs} . I provide three constructors—operational recipes for building F_{obs} from available data—ordered from most data-rich to most generic.

¹The Cramér distance is closely related to the energy distance of Székely and Rizzo (2013): $\int (F-G)^2 dz = \frac{1}{2} \varepsilon(F, G)$ in the univariate case.

Constructor 1: Dataset quality bounds

Many event datasets provide explicit low/best/high estimates per observation. The UCDP Georeferenced Event Dataset (Uppsala Conflict Data Program, 2025), for instance, records best, low, and high fatality estimates derived from the coding process—these bounds represent coder disagreement about the plausible range, not quantiles of a measurement-error distribution. I treat them pragmatically as the support of either a uniform or a log-normal distribution centred on the best estimate:

$$F_{\text{obs}}(z) = \text{Uniform}(z; \text{low}, \text{high}) \quad \text{or} \quad F_{\text{obs}}(z) = \text{LogNormal}(z; \log(\text{best}), \sigma_{\text{LN}}) \quad (4)$$

where σ_{LN} is set so that approximately 95% of mass falls in $[\text{low}, \text{high}]$. This is the most principled constructor because it uses the dataset’s own quality metadata, external to the forecaster being evaluated.

Constructor 2: RVI Gumbel mixture

Vesco et al. (2026) asked conflict-data coders to estimate plausible ranges for a set of reported fatality counts. The responses revealed a consistent pattern: some fraction of coders effectively reproduced the reported value, while the remainder believed the true count was likely higher. Vesco et al. capture this in a Reported-Value Inflated (RVI) Gumbel mixture model, where the Gumbel component is a right-skewed distribution that places more probability mass above the reported value than below—matching the asymmetric direction of conflict-fatality underreporting. The RVI CDF is

$$F_{\text{obs}}(z) = w \cdot \mathbf{1}(z \geq \tilde{y}) + (1 - w) \cdot \text{Gumbel}(z; \mu_g, \beta_g), \quad (5)$$

where w is the probability mass on the reported value $\tilde{y}$, and (μ_g, β_g) are the Gumbel location and scale. The location and scale parameters (μ_g, β_g) are predicted from $\log(1 + \tilde{y})$ and vague-number indicator dummies via linear regression; w is predicted from the same covariates via logistic regression. I use the coefficient estimates from Table A10 in Vesco et al. (2026), which pools across violence types (36 coefficients total: intercept + slope for each of w , μ_g , β_g , plus vague-number terms).² This constructor captures the asymmetric underreporting that symmetric parametric constructors miss: for low-fatality events, where underreporting is most severe, the Gumbel tail spreads substantially above the reported count. The vague-number indicator dummies are designed for event-level reported values; applying them to cell-month aggregates is a known approximation (see §5 for discussion). Where UCDP data are available, this constructor supersedes the parametric default.

²The implementation clamps $w \in [0.01, 0.99]$ and $\mu_g \geq 0$ to prevent degenerate predictions; the Gumbel CDF assigns non-zero probability to negative fatalities, which is handled by truncating at zero.

Constructor 3: Parametric default

For datasets without explicit bounds or an empirical uncertainty model, a default is a log-normal centred on the reported value with a fixed relative uncertainty:

$$F_{\text{obs}}(z) = \text{LogNormal}(z; \log(y), \sigma_{\text{rel}}). \quad (6)$$

The parameter σ_{rel} should be informed by domain knowledge—the typical revision rate between successive dataset releases, the known underreporting ratio, or expert elicitation. For armed-conflict fatality counts, I use $\sigma_{\text{rel}} = 0.4$ as a working default. In practice, this places roughly two-thirds of the observational distribution between $0.67y$ and $1.49y$ —for a reported count of 50 deaths, the plausible range runs from about 34 to 75. The RVI Gumbel model of Vesco et al. (2026) (Constructor 2) implies that the effective log-space standard deviation of observational uncertainty varies from approximately 0.2 for large events ($\tilde{y} > 100$) to over 0.9 for single-fatality events; $\sigma_{\text{rel}} = 0.4$ approximates the range typical of small-to-medium events. Where event-specific uncertainty is available, Constructor 2 supersedes this parametric default. A community using this reformulation for benchmarking would need to agree on standard σ_{rel} values per data type, analogous to standardised baselines in forecast competitions.

Numerical implementation. The integral in Eq. 3 is computed by trapezoidal integration over a uniform grid of 4,001 points covering the support of both F_{pred} and F_{obs} . Forecast CDFs are represented as empirical CDFs from 4,000 Monte Carlo samples (random draws used to approximate the CDF numerically; seed = 0 throughout). Doubling the grid to 8,001 points changes scores by $< 0.01\%$; doubling the Monte Carlo sample to 8,000 changes individual scores by up to $\sim 4\%$ but does not affect rankings. For specific parametric families where both CDFs are log-normal, closed-form expressions for the IQD exist; the reference implementation uses the general numerical form to accommodate arbitrary F_{obs} constructors.

4 Consistency Properties

The Cramér distance between two fixed CDFs is a proper divergence: it is non-negative and equals zero if and only if the two distributions are identical (Székely and Rizzo, 2013; Thorarinsdottir et al., 2013). This is a weaker guarantee than the classical propriety of §2—it says that the best forecast *for a given observation* is the observation itself, but does not directly address what happens when the observation is stochastic.

The substantive question is whether the pipeline—constructing F_{obs} from a noisy draw y and scoring F_{pred} against it—rewards forecasts that get the location right. I verify this numerically: drawing $N = 500$ observations from a known distribution, constructing F_{obs} for each draw, and sweeping a forecast parameter μ across a grid. The argmin falls at the true μ for all tested values across three distributional families (log-normal, Gaussian, count-like; §5). A formal analytical proof via the energy-distance framework is deferred to a companion paper.

Scale-consistency does *not* hold: the minimising spread absorbs the observation noise, just

as a blurred photograph combines camera-shake and lens-blur into a single effective width. For conflict fatality data, this effect is more consequential than in meteorology or hydrology: the RVI model of Vesco et al. (2026) implies observational uncertainty of $\sigma_{\text{rel}} \approx 0.2\text{--}0.9$ in log-space, which can equal or exceed forecast-model spread. The metric rewards forecasts that match the observable data distribution, not the latent truth—the same property classical CRPS has in degenerate form. This does not affect the ranking reversals demonstrated in §5, which depend on location-consistency rather than scale-consistency. Formal details—including the L^2 minimisation proof, the stochastic-observation consistency definition, numerical verification, and the full scale-consistency analysis—are in Appendix A.

5 Demonstration

5.1 Controlled demonstration

Setup. Does accounting for observational uncertainty change which forecast version wins? I start with a controlled case: both versions are centred on the reported value y , which isolates the spread dimension—both metrics handle location identically. The setup:

- The reported observation is $y = 50$.
- The assumed observational uncertainty is $\sigma_{\text{rel}} = 0.4$ in log-space (the Constructor 3 working default; see §3 for calibration against the RVI Gumbel model).
- Two forecast versions predict the same observation:
 - **Tight (overconfident):** $\text{LogNormal}(\log y, \sigma_{\text{tight}} = 0.05)$ —concentrates nearly all probability mass on a single value, claiming to know the truth precisely.
 - **Honest:** $\text{LogNormal}(\log y, \sigma_{\text{honest}} = 0.4)$ —a wider distribution matching the true observational uncertainty.

Results. Classical CRPS ranks the tight version first: its narrow distribution is closer to the step at y at almost every z , and the squared-area integrand is smaller. The reformulated metric, computed against a smooth F_{obs} constructed via Constructor 3 ($\sigma_{\text{rel}} = 0.4$), reverses the ranking: the honest version wins. Figure 2 shows the four scores. The reversal is qualitative, not a numerical artifact. In this baseline case, the reversal is algebraically guaranteed: the Cramér distance is an L^2 metric, and $\sigma_{\text{honest}} = \sigma_{\text{rel}}$ means the honest version’s CDF matches F_{obs} exactly, which is the minimiser by construction. The substantive question is not *whether* the reversal occurs at the matched parameterisation, but *how far* the honest version can deviate from F_{obs} and still beat the overconfident version.

The reversal does not depend on the honest version exactly matching F_{obs} : varying σ_{honest} while holding $\sigma_{\text{rel}} = 0.4$ fixed, the reversal holds for $\sigma_{\text{honest}} \in [0.10, 0.85]$ —the honest version can be $4\times$ too narrow or $2\times$ too wide and still beat the tight version. The reversal disappears when observational uncertainty is small ($\sigma_{\text{rel}} \leq 0.10$) or when the honest version is extremely wide

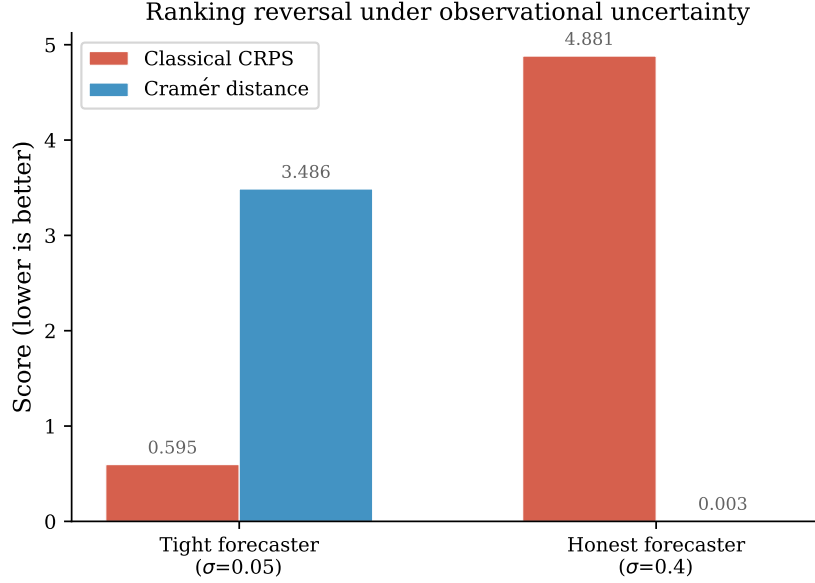


Figure 2: Ranking reversal between two forecast versions on a controlled example. Under classical CRPS the tight (overconfident) version wins; under the Cramér-distance reformulation the honest version wins. The reversal is qualitative—the sign of which version is better changes.

($\sigma_{\text{honest}} \geq 1.0$). Detailed robustness analysis—including constructor comparison, location-error tolerance, and generalisation across distributional families—is in Appendix B.

5.2 Empirical comparison on UCDP conflict data

Data and design. I replicate the controlled demonstration on real observations from the UCDP Georeferenced Event Dataset (GED, data version 24.1, final-coded events; codebook: Uppsala Conflict Data Program 2025), using best-estimate fatality counts from 2020–2024 aggregated to the PRIO-GRID cell-month level (a $0.5^\circ \times 0.5^\circ$ geographic grid cell observed in a single calendar month, the standard spatiotemporal unit in VIEWS-format evaluations). I pool state-based, non-state, and one-sided violence and retain cell-months with at least one recorded fatality; historical uncertainty is derived from cells with ≥ 3 active months during 1989–2019. This yields 19,397 cell-months from 2,065 cells. This sample covers the active-conflict portion of the evaluation space; the roughly 800,000 cell-months with zero recorded fatalities are excluded because F_{obs} degenerates to a step function at zero and the reformulation reduces to classical CRPS. The results therefore characterise the regime where observational uncertainty is operative, not the full VIEWS evaluation grid.

For each cell-month I construct two forecast versions, both centred on the reported best estimate y :

- **Tight:** $\text{LogNormal}(\log y, 0.05)$ —claims to know the observation precisely.
- **Honest:** $\text{LogNormal}(\log y, \hat{\sigma}_{\text{cell}})$ —where $\hat{\sigma}_{\text{cell}} = \sqrt{\log(1 + \text{CV}^2)}$ is derived from the cell’s historical coefficient of variation. Median $\hat{\sigma}_{\text{cell}} = 0.97$; interquartile range $[0.81, 1.16]$; clipped to $[0.1, 2.0]$ (affecting 1.1% of cell-months).

I score both under classical CRPS and the Cramér-distance reformulation with Constructor 3 ($\sigma_{\text{rel}} = 0.4$).

Results. Under classical CRPS the tight version wins in 100% of cell-months—a direct consequence of both versions being centred on y and CRPS’s well-known preference for concentration. Under the Cramér-distance reformulation, the ranking reverses in 39% of cell-months overall.³ The reversal rate depends sharply on forecast uncertainty: for cells with moderate uncertainty ($\hat{\sigma}_{\text{cell}} \leq 0.8$; $n = 4,716$), the honest version wins in *every* cell-month. For cells with very high uncertainty ($\hat{\sigma}_{\text{cell}} > 1.0$; $n = 8,902$), the honest version is too wide and loses—exactly the regime identified in the controlled demonstration. Figure 3 shows the reversal rate by uncertainty regime and the score-difference distribution.

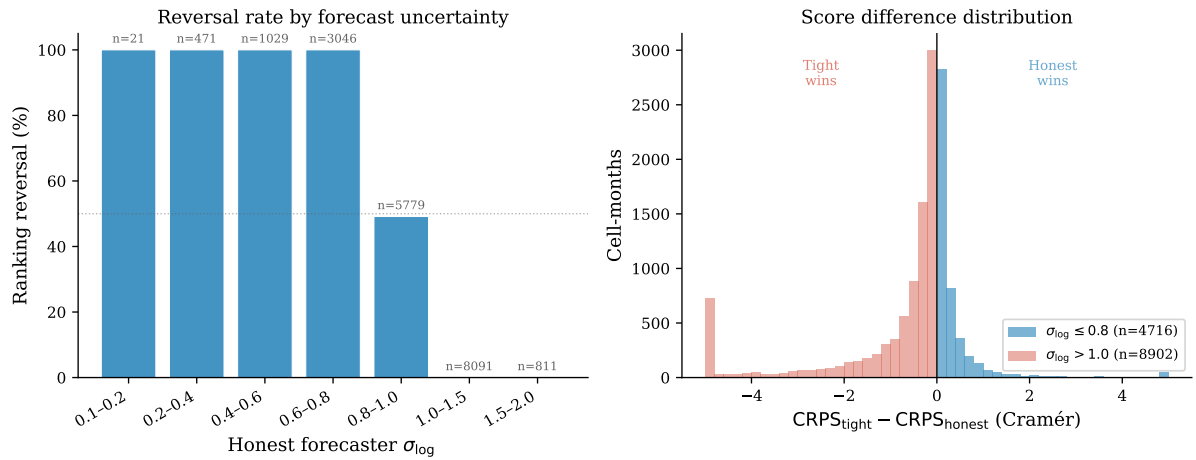


Figure 3: Ranking reversal on UCDP conflict data (2020–2024, 19,397 cell-months). Left: reversal rate under the Cramér-distance reformulation as a function of the honest version’s log-space uncertainty $\hat{\sigma}_{\text{cell}}$. The reversal is complete for $\hat{\sigma}_{\text{cell}} \leq 0.8$ and absent for $\hat{\sigma}_{\text{cell}} > 1.0$. Right: distribution of score differences; positive values indicate the honest version wins.

Among cell-months with genuine UCDP low/best/high bounds, Constructor 1 produces a reversal rate of only 9.9%—reflecting the fact that UCDP’s coding ranges are narrow relative to the true observational uncertainty (Vesco et al., 2026). Under the RVI Gumbel mixture (Constructor 2), the reversal widens to 80.8% of cell-months, capturing asymmetric underreporting that the symmetric parametric constructor misses. The Bessac–Naveau conditional-expectation correction (Bessac and Naveau, 2021) confirms the pattern, with 98.1% concordance overall. A per-violence-type breakdown, constructor comparison details, and the unit-of-analysis caveat for Constructor 2 are in Appendix B.

5.3 Validation with an operational forecaster

The preceding comparisons centre both versions on the reported value y —information unavailable at prediction time. This isolates the spread effect but leaves open whether the reversal survives when the model does not observe y . I test this using HydraNet (von der Maase, 2025), a spatio-temporal Monte Carlo Dropout architecture trained on the Violence & Impacts Early-Warning

³The reversal rate is sensitive to σ_{rel} : at $\sigma_{\text{rel}} = 0.3$ it drops to 12.6%; at $\sigma_{\text{rel}} = 0.5$ it rises to 67.7%.

System (VIEWS) PRIO-GRID. The model produces 64 posterior samples per cell-month⁴ for each violence type; I use state-based fatalities, the best-documented violence type with the lowest reversal rate in the synthetic comparison (79.3% under RVI; Appendix B)—the reformulation’s impact is likely larger for non-state and one-sided violence, where observational uncertainty is greater.

Setup. The model is trained on January 1990–December 2020 and produces 36-step-ahead validation predictions for January 2021–December 2024 (the standard VIEWS validation partition). I match these against UCDP state-based fatality actuals aggregated to the cell-month level, using the shortest-horizon (most recent) prediction available for each cell-month, yielding 6,725 cell-months across 1,173 cells where the reported best estimate is positive. For each cell-month, I construct two versions of the same model’s output—this tests whether the metric values uncertainty quantification, not whether it changes model selection between genuinely different systems:

- **Full posterior:** the model’s 64 posterior samples, used directly as the forecast distribution.
- **Tight:** $\text{LogNormal}(\log \bar{x}, 0.05)$ where $\bar{x}$ is the posterior mean—a version that retains the model’s location estimate but discards its uncertainty quantification.

Both versions share the same location; the only difference is spread.

Results. In the full sample, classical CRPS strictly prefers the tight version in only 2.5% of cell-months (167/6,725). A further 16.4% (1,102 cell-months) are ties where the model predicts zero fatalities and both versions are degenerate; the remaining 81.1% strictly prefer the full posterior. The dominance of the full posterior reflects HydraNet’s typically large location errors; when the model misses the location, wider spread helps under classical CRPS simply by placing more mass near the target.

The interesting regime is the 1,031 cell-months (15.3%) where the model’s prediction falls within a factor of two of the actual—where location error is moderate and the spread dimension becomes consequential. In this subset, classical CRPS selects the tight version in 167 cell-months (16.2%). The Cramér-distance reformulation (Constructor 3, $\sigma_{\text{rel}} = 0.4$) reverses 166 of these 167 cases (99.4%), selecting the full posterior instead. Under the RVI Gumbel constructor (Constructor 2), the reversal rate is 167/167 (100%). The single cell-month that resists reversal under Constructor 3 is a high-fatality case (best = 56) where the full posterior has a log-space standard deviation of 0.77—genuinely too wide relative to F_{obs} . The Cramér distance does not reward uncertainty indiscriminately but penalises spread that exceeds observational uncertainty, exactly as in the synthetic demonstration.

⁴The HydraNet architecture draws 128 samples per cell-month in its reference configuration; the validation here uses 64 due to local memory constraints.

6 Discussion

Choosing F_{obs} in practice. The three constructors have different epistemic credentials. Dataset quality bounds (Constructor 1) are the most defensible because they are external to the forecaster being evaluated; the low 9.9% reversal rate under Constructor 1 (§5) itself suggests that UCDP coding ranges understate true observational uncertainty. Parametric defaults (Constructor 3) are practical but introduce a free parameter; a careless choice of σ_{rel} can mask or invert the ranking, and benchmarking communities would need to agree on standard values per data type. The RVI Gumbel mixture (Constructor 2) bridges these approaches: it uses regression coefficients estimated from coder surveys (Vesco et al., 2026) to predict the distribution of plausible fatality counts, capturing the asymmetric underreporting that symmetric parametric constructors miss. The Vesco et al. model has its own limitations—response biases in coder surveys, assumed distributional form, limited sample size—and should be read as a principled estimate rather than ground truth. That said, I recommend Constructor 2 where UCDP data are available, Constructor 1 where quality bounds exist for other datasets, and Constructor 3 with a community-agreed σ_{rel} otherwise. A caveat: the Constructor 2 results in §5 apply the RVI regression at the cell-month aggregate level rather than per event, introducing a unit-of-analysis mismatch that likely understates the true observational uncertainty (see §5); event-level application would be more principled but requires combining event-level distributions. A fourth possibility—deriving F_{obs} from a model’s own confidence signal—is conceptually available but introduces circularity; I do not pursue it here.

Relation to prior approaches. The problem of evaluating forecasts against imprecise observations is well studied; the present work applies ideas developed primarily in meteorology to conflict forecasting. One structural consequence worth noting first: Bröcker (2009) decomposes proper scores into reliability, resolution, and uncertainty components; the Cramér-distance reformulation shifts mass from the “uncertainty” term—which in the classical decomposition absorbs all observational noise—into the smooth F_{obs} in the integrand. A formal restatement of this decomposition for the reformulated metric is deferred to a technical companion.

Scoring under observation error. Bröcker and Smith (2007) highlight the need for generalised scores when verification data are uncertain, proposing a convolution approach: the forecast is convolved with the observation-error kernel before scoring, so that forecast and observation are compared on equal footing. Bessac and Naveau (2021) take a different route, computing $\mathbb{E}[S(F, Y) | \tilde{y}]$ —the conditional expectation of the score over the observation error distribution—yielding an unbiased estimate of the true score. This conditional-expectation correction is principled when the observation error model is known. The Cramér-distance reformulation takes a third route: rather than correcting the score after the fact, it replaces the degenerate observation CDF with a smooth F_{obs} *inside the integrand*, changing what is being measured. The Bröcker–Smith convolution asks “what would the score look like if the forecast absorbed the observation noise?”; the Bessac–Naveau correction asks “what would the score have been had we observed the truth?”; the Cramér-distance reformulation asks “what is the distance between the forecast distribution and the observation distribution?” The three approaches yield the same

ranking in expectation when F_{obs} is correctly specified, but they differ in what they require: the convolution approach requires a noise kernel, the conditional-expectation approach requires a tractable posterior $[X | Y]$, and the CDF-substitution approach requires a constructable F_{obs} .

I verify this empirically by comparing the Cramér-distance reformulation against the Bessac–Naveau conditional-expectation correction. On the controlled demonstration (§5), both approaches select the same winner (honest version) for all $\sigma_{\text{honest}} \leq 0.80$ and agree that the tight version wins for $\sigma_{\text{honest}} \geq 0.90$. A narrow disagreement band exists at $\sigma_{\text{honest}} \approx 0.85\text{--}0.88$, where the Cramér distance still selects the honest version but the conditional-expectation correction flips to tight—the CDF-substitution approach is slightly more tolerant of wide versions near the transition. On the UCDP empirical comparison (19,397 cell-months, Constructor 3 with $\sigma_{\text{rel}} = 0.4$), the two approaches agree in 98.1% of cell-months overall. Outside the transition zone ($\hat{\sigma}_{\text{cell}} \leq 0.8$ or > 1.0) concordance is 100%; within the transition zone ($0.8 < \hat{\sigma}_{\text{cell}} \leq 1.0$) it drops to 93.6%—roughly one in fifteen cell-months receives a different ranking depending on the method (§5). The residual disagreement is concentrated precisely where the decision is hardest: where observational uncertainty is large enough to be consequential but not so large as to swamp the forecast signal. In this disagreement band, the conditional-expectation correction has the stronger theoretical warrant—it recovers an unbiased estimate of the true score under correct error-model specification—whereas the Cramér distance measures a different quantity (distribution-to-distribution distance) whose optimality depends on the F_{obs} constructor.

Error-corrected and distributional approaches. Ferro (2017b) constructs error-corrected proper scoring rules whose expected values, taken over the observation-error distribution, equal the original proper score evaluated at the true value—a frequentist unbiasedness condition that requires an explicit error model but not a Bayesian posterior. Candille and Talagrand (2008) study the impact of observation error on ensemble verification diagnostics, proposing an “observational-probability” method that replaces point observations with probability distributions in scores such as the Brier score and reliability diagrams. Bessac and Naveau (2021) derive the full distribution of CRPS under Gaussian additive and multiplicative-Gamma observation errors, showing that imperfect verification data cause systematic underdispersion bias—an effect closely related to the ranking reversal demonstrated in §5.

The underdispersion bias Bessac and Naveau predict analytically is directly observable in the UCDP data (§5): classical CRPS selects the tight version in 100% of cell-months, and the Cramér-distance reformulation reverses this in precisely the regime where observational uncertainty exceeds forecast uncertainty (Spearman $\rho = -0.38$, $p < 10^{-15}$, between σ_{honest} and the classical score gap).

These approaches share the same motivation as the present work but address different aspects: score bias (Bessac–Naveau), error-corrected scoring (Ferro), or diagnostic calibration (Candille–Talagrand), whereas I focus on the effect on model ranking.

Scale-sensitivity and ensemble size. Bolin and Wallin (2023) show that CRPS is not locally scale-invariant: it overweights observations in regions where the forecast distribution has large variability. They propose a Scaled CRPS that normalises by local spread—a complementary

approach that could in principle be combined with the reformulation proposed here. Ferro (2017a) addresses finite ensemble size bias, a distinct but related source of overconfidence in CRPS rankings. Weijs et al. (2011) cast the problem in information-theoretic terms, proposing divergence-based scores with explicit uncertainty decompositions; their framework is more general but requires different scoring infrastructure.

Among these approaches, the Cramér-distance reformulation is attractive for domains where observation error models are not well characterised: it requires only a constructable F_{obs} , not an invertible error model, and reduces exactly to classical CRPS when observations are precise. Its main advantage for conflict forecasting is that it provides a natural entry point for empirically grounded uncertainty models: UCDP’s own low/best/high bounds are coding artefacts that understate the true observational uncertainty (§5), but the RVI Gumbel mixture of Vesco et al. (2026)—built precisely because those bounds are insufficient—can be plugged directly into the F_{obs} constructor. Fields with well-characterised instrument error covariances (e.g. numerical weather prediction) may benefit more from the conditional-expectation correction of Bessac and Naveau (2021), the convolution approach of Bröcker and Smith (2007), or the error-corrected scoring rules of Ferro (2017b).

What the reformulation does not solve.

- **It does not solve the Forecaster’s Dilemma.** The result of Lerch et al. (2017)—that conditioning forecast evaluation on extreme observations systematically discredits skillful forecasts—holds for the reformulated metric just as for classical CRPS. Weighted scoring rules that emphasise tail behaviour (Gneiting and Ranjan, 2011) are still needed as complementary diagnostics.
- **It does not eliminate distributional assumptions.** The constructor of F_{obs} is itself a distributional assumption. The reformulation moves the assumption from “observations are exact” to “observations follow this distribution”—more honest, but still an assumption open to mis-specification.
- **Additional sources of observational uncertainty are unaddressed.** Systematic data-quality biases, spatial variation in reporting quality, georeferencing imprecision (UCDP precision codes 1–7), and false-negative uncertainty in the roughly 800,000 zero-fatality cell-months excluded from §5 each represent distinct error sources not captured by F_{obs} .
- **The demonstrations use oracle-centred forecast versions.** The controlled and UCDP comparisons centre both versions on the reported value y —information unavailable at prediction time. The 39% reversal rate describes a regime of perfect location knowledge; the HydraNet validation (§5.3) tests whether the reversal survives genuine location error. All three comparisons test uncertainty valuation rather than model selection—whether the reformulation changes rankings between genuinely different models is untested.

Sensitivity to F_{obs} mis-specification. Could the metric’s behaviour under a mis-specified F_{obs} be worse than classical CRPS with no F_{obs} at all? I quantify this by running the stochastic

consistency check from §4 (details in Appendix A) under deliberate mis-specification. The true data-generating distribution is $\text{LogNormal}(\mu = 1.0, \sigma = 0.3)$; I construct F_{obs} using σ_{rel} values ranging from 0.1 to 1.0 and ask where the expected Cramér distance is minimised.

Table 1 shows the results. When σ_{rel} matches the true σ (ratio 1.0 $\times$), the argmin falls exactly on μ_{true} . Moderate overestimation (up to 2 $\times$) shifts the argmin by at most one grid step (+0.05). Severe overestimation (3.3 $\times$) shifts it by +0.15—meaningful but bounded. Underestimation produces no detectable bias in this sweep. For this log-normal DGP, the bias is monotonic in the mis-specification ratio; whether monotonicity holds across distributional families is untested, but the practical implication—a practitioner can bound the bias by bounding the mis-specification—applies whenever the relationship is monotone.

The asymmetry has a structural explanation. When σ_{rel} is underestimated, F_{obs} approaches the degenerate step function at y , and the Cramér distance reduces to classical CRPS—which is proper, so the location argmin remains at μ_{true} . When σ_{rel} is overestimated, F_{obs} is wider than the true observation noise; the log-normal right-skew of the widened F_{obs} shifts the L^2 -minimising location upward, producing positive bias. The same mechanism explains the scale-consistency convolution effect discussed in §4 and Appendix A: the metric measures distance from the observable distribution, which inherits the observation noise.

A formal characterisation of this bias is an open problem.

σ_{rel}	Ratio ($\sigma_{\text{rel}}/\sigma_{\text{true}}$)	Argmin $\hat{\mu}$	Bias ($\hat{\mu} - \mu_{\text{true}}$)
0.1	0.3 $\times$	1.000	0.000
0.2	0.7 $\times$	1.000	0.000
0.3	1.0 $\times$	1.000	0.000
0.4	1.3 $\times$	1.050	+0.050
0.6	2.0 $\times$	1.050	+0.050
0.8	2.7 $\times$	1.100	+0.100
1.0	3.3 $\times$	1.150	+0.150

Table 1: Argmin bias under mis-specified F_{obs} . The true DGP is $\text{LogNormal}(1.0, 0.3)$; F_{obs} uses varying σ_{rel} . Monte Carlo expectation over 500 draws. Grid step = 0.05.

Implications for forecast competitions. Forecast competitions and model selection procedures that use CRPS as the sole ranking metric implicitly assume observations are exact. When this assumption is violated, competition rankings can systematically favour overconfident models. The practical recommendation is not to replace CRPS with the Cramér-distance reformulation in all settings—classical CRPS is exactly right when observations are precise—but to use the reformulation as a secondary metric or robustness check in domains where observational uncertainty is known to be substantial. This recommendation rests on location-consistency verified numerically across three distributional families (§5; Appendix B, Table 4). Domains where this is immediately relevant include conflict forecasting (Hegre et al., 2019; Uppsala Conflict Data Program, 2025), epidemiological surveillance (Gibbons et al., 2014), and economic forecasting from survey data.

Open questions.

- Closed-form expressions for the reformulated metric when both F_{pred} and F_{obs} belong to common parametric families (log-normal, gamma, Weibull) would speed evaluation and enable analytical optimisation.
- A formal Bröcker decomposition of the reformulated metric into reliability, resolution, and observational-uncertainty components.
- An analytical proof of consistency under stochastic observations via the energy-distance framework, with explicit conditions on the F_{obs} constructor.
- Empirical comparison of model rankings on operational benchmarks (conflict forecasting, weather prediction, epidemiological surveillance) under classical and reformulated CRPS with each constructor.
- Convergence analysis: how does the bias from a mis-specified F_{obs} scale with the magnitude and direction of the mis-specification?

A Consistency Properties: Formal Details

L^2 minimisation (fixed F_{obs}). For any fixed observational CDF F_{obs} , the integral $\int (F_{\text{pred}}(z) - F_{\text{obs}}(z))^2 dz$ is non-negative and equals zero if and only if $F_{\text{pred}}(z) = F_{\text{obs}}(z)$ for all z . The minimiser over deterministic F_{pred} is therefore F_{obs} itself. This is an L^2 minimisation statement, not a propriety theorem in the Gneiting and Raftery (2007) sense—classical propriety requires an expectation over the random observation y , which the fixed- F_{obs} statement does not involve.

Consistency under stochastic observations. Write $F_{\text{obs}}(y)$ for the observational CDF built from a specific draw y . The relevant quantity is

$$\mathbb{E}_{y \sim Q} [\text{CRPS}_{\text{obs}}(F_{\text{pred}}, F_{\text{obs}}(y))]. \quad (7)$$

The metric should reward forecasts that get the location right, even when individual observations are noisy. Formally, I say that the pipeline is *location-consistent* at $F_{\text{pred}} = Q$ if this expectation is minimised over the forecast location parameter when F_{pred} equals the true data-generating distribution Q . Consistency in this sense is weaker than classical propriety—it depends on the F_{obs} constructor being well-specified with respect to Q —but it is the property a practitioner needs: reporting the true location should yield the best expected score.

The honesty of the metric depends on the honesty of the F_{obs} constructor. A practitioner who specifies $\sigma_{\text{rel}} = 0.3$ when the true relative uncertainty is 0.1 will get a metric that rewards forecasts that are too wide; one who specifies σ_{rel} too small will get a metric that rewards overconfidence. The reformulation does not eliminate the distributional assumption—it moves it from inside the score formula to the constructor of F_{obs} , where it is visible, auditable, and open to sensitivity analysis.

Numerical consistency check. I verify consistency by drawing $N = 500$ observations $y^{(1)}, \dots, y^{(N)}$ from a known log-normal truth with parameters μ_{true} and $\sigma = 0.3$ (a test parameter

chosen for the numerical check; the operational default for conflict data is $\sigma_{\text{rel}} = 0.4$, §3), constructing $F_{\text{obs}}(y^{(i)})$ parametrically for each draw with $\sigma_{\text{rel}} = 0.3$, sweeping a forecast parameter μ across a grid, and computing the expected score $\frac{1}{N} \sum_i \text{CRPS}_{\text{obs}}(F_{\text{pred}}(\mu), F_{\text{obs}}(y^{(i)}))$ for each μ . The argmin should fall near μ_{true} .

Figure 4 shows the result for $\mu_{\text{true}} = 1.0$. At grid step $\Delta\mu = 0.05$ ($N = 500$ draws), the argmin falls on $\mu = 1.0$ for all four tested values $\mu_{\text{true}} \in \{0.5, 1.0, 1.5, 2.0\}$. Refining to $\Delta\mu = 0.005$ with $N = 2,000$ draws, the residual bias is ≤ 0.01 and varies by seed—consistent with Monte Carlo noise at the grid resolution rather than a systematic shift.

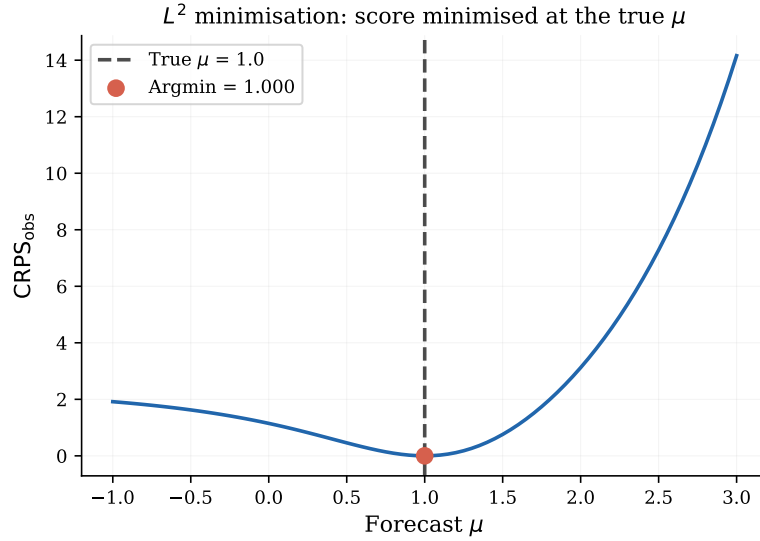


Figure 4: Location-consistency check. The expected Cramér distance, averaged over $N = 500$ stochastic draws of y from the true distribution (each with its own $F_{\text{obs}}(y^{(i)})$), is minimised at the true μ .

Scale consistency: a convolution effect. The μ -sweep above verifies location-consistency. A natural follow-up is whether the metric is also minimised at the true scale parameter σ_{true} . It is not—and the reason is intuitive: the metric rewards forecasts that match the truth *as it would appear through the observation process*. Just as a blurred photograph combines camera-shake and lens-blur so that the result is wider than either source alone, the metric combines forecast spread and observation noise into a single effective width. For $\text{LogNormal}(\mu = 1.0, \sigma_{\text{true}} = 0.3)$ with observation-noise parameter $\sigma_{\text{obs}} = 0.3$ (equal to σ_{rel} in the well-specified case), the expected Cramér distance is minimised at $\sigma_{\text{pred}} \approx 0.42$, not 0.30. Under a Gaussian DGP ($\mu = 50$, $\sigma_{\text{true}} = 15$, $\sigma_{\text{obs}} = 10$), the argmin falls at $\sigma_{\text{pred}} \approx 18 \approx \sqrt{15^2 + 10^2}$. In both cases the minimiser is the convolution width $\sqrt{\sigma_{\text{true}}^2 + \sigma_{\text{obs}}^2}$: the expectation over stochastic y effectively convolves F_{pred} with the observation noise. For conflict fatality data, this effect is more consequential than in meteorology or hydrology: the RVI model of Vesco et al. (2026) implies observational uncertainty of $\sigma_{\text{rel}} \approx 0.2$ – 0.9 in log-space, which can equal or exceed forecast-model spread. When $\sigma_{\text{obs}} \geq \sigma_{\text{true}}$, the convolution width is dominated by observation noise, and the scale optimum shifts substantially from the true forecast spread.

The observable distribution is $Q * F_{\varepsilon}$, not the latent Q itself. Classical CRPS has the same

property in degenerate form: when $F_{\text{obs}} = \delta_y$ and $y = \theta + \varepsilon$, the CRPS-optimal forecast is $Q * F_\varepsilon$, not Q . Consistency therefore holds for location but not for scale: the metric rewards forecasts that match the observable data distribution, penalising overconfidence relative to what the data can actually resolve.

B Supplementary Demonstration Results

B.1 Constructor comparison

The controlled demonstration in §5.1 uses Constructor 3 (parametric). Table 2 repeats the comparison with Constructor 1 (dataset quality bounds), using UCDP-style low/best/high bounds around the same $y = 50$. With wide bounds $[25, 50, 100]$ —typical of events in regions with limited media coverage—the ranking reversal holds: the honest version wins. With narrow bounds $[45, 50, 55]$ —typical of well-documented events—the reversal disappears and the tight version wins, which is correct: when the observation is known precisely, a tight forecast is the right answer.

F_{obs} constructor	Parameters	Tight	Honest	Winner
Classical CRPS (step at y)	—	0.595	4.881	Tight
Constructor 3 (parametric)	$\sigma_{\text{rel}} = 0.4$	3.486	0.003	Honest
Constructor 1 (bounds)	$[25, 50, 100]$	6.997	1.361	Honest
Constructor 1 (bounds)	$[45, 50, 55]$	0.040	3.373	Tight

Table 2: Scores for the tight and honest forecast versions under classical CRPS, the parametric Cramér-distance reformulation (Constructor 3), and the bounds-based reformulation (Constructor 1). The ranking reversal occurs when observational uncertainty is substantial—either via a wide σ_{rel} or via wide low/high bounds—and disappears when bounds are narrow. The parametric constructor uses $\sigma_{\text{rel}} = 0.4$ (the working default defined in §3).

B.2 Robustness and boundary conditions

Boundaries. The reversal does not hold universally. It disappears when $\sigma_{\text{rel}} \leq 0.10$ (observational uncertainty is small enough that a tight forecast is genuinely correct), when $\sigma_{\text{tight}} \geq \sigma_{\text{rel}}$ (the “tight” version is no longer meaningfully tight), or when $\sigma_{\text{honest}} \geq 1.0$ (the honest version is so wide it no longer overlaps F_{obs}). The reformulation changes the ranking only when observational uncertainty is large enough to be consequential.

A natural concern is that the reversal depends on the honest version’s spread exactly matching F_{obs} . It does not. Varying σ_{honest} while holding $\sigma_{\text{rel}} = 0.4$ fixed, the reversal holds for $\sigma_{\text{honest}} \in [0.10, 0.85]$ —the honest version can be $4\times$ too narrow or $2\times$ too wide and still beat the tight version. The crossover occurs at $\sigma_{\text{honest}} \approx 0.88$. This robustness arises because the reversal is driven by the tight version’s *failure*: its extremely narrow CDF ($\sigma = 0.05$) is far from the smooth F_{obs} at almost every evaluation point, producing a Cramér-distance score of approximately 3.49 regardless of what the honest version does. The honest version’s score varies from 0.003 (when matching F_{obs}) to 3.1 (at $\sigma_{\text{honest}} = 0.85$), staying below 3.49 across the entire range.

Location error tolerance. The controlled experiments centre both versions on the reported value y . What happens when the honest version also has location error? I test this by shifting the honest version’s location to $y \cdot e^\delta$ in the original space (equivalently, adding δ in log-space) while keeping the tight version centred at y and holding $\sigma_{\text{rel}} = 0.4$ fixed. Table 3 shows the results.

δ	Honest centre	Tight	Honest	Winner	Location error
0.00	50.0	3.486	0.003	Honest	0%
0.10	55.3	3.486	0.449	Honest	+11%
0.20	61.1	3.486	1.736	Honest	+22%
0.28	66.2	3.486	3.432	Honest	+32%
0.29	66.8	3.486	3.687	Tight	+34%
0.40	74.6	3.486	7.175	Tight	+49%

Table 3: Cramér-distance scores under location error. The honest version ($\sigma_{\text{honest}} = 0.4$) is shifted δ units in log-space from the reported value $y = 50$; the tight version ($\sigma_{\text{tight}} = 0.05$) remains centred at y . The reversal persists until the honest version’s location error exceeds roughly 33% of the reported value, at which point location error dominates spread calibration.

The reversal persists for location errors up to $\delta \approx 0.28$ (a 33% shift in the original space). This tolerance is substantial: an honest version that is one-third off in location but correctly calibrated for spread still beats a tight version that nails the location but ignores observational uncertainty. The crossover at $\delta \approx 0.29$ occurs where the honest version’s location error penalty exceeds the tight version’s fixed spread penalty (≈ 3.49). Beyond this point, the tight version wins under both metrics.

Generalisation across distributional families. The controlled demonstration uses log-normal distributions throughout. To confirm that the propriety and ranking-reversal results are not artifacts of this choice, I repeat the analysis under two additional data-generating processes: (i) a Gaussian DGP with $y \sim \text{Normal}(50, 15)$, Gaussian F_{obs} ($\sigma_{\text{obs}} = 10$), and Gaussian forecast versions (tight $\sigma = 2$, honest $\sigma = 10$); and (ii) a count-like regime representative of conflict data ($y \in \{3, 5, 10, 20\}$, $\sigma_{\text{rel}} = 0.4$), using the same log-normal parameterisation for F_{obs} and both versions. Table 4 summarises the results.

DGP	Parameters	Location bias	Reversal range	BN concordance
LogNormal	$\mu_{\log} = 1.0, \sigma = 0.3$	0.000	$\sigma_h \in [0.10, 0.85]$	agree
Normal	$\mu = 50, \sigma = 10$	0.000	$\sigma_h \in [2.5, 20.0]$	agree
Count-like	$y \in \{3, 5, 10, 20\}$	0.000	all $y > 0$ reversed	agree

Table 4: Propriety and ranking-reversal results across three distributional families. “Location bias” is the distance between the μ -sweep argmin and the true location parameter (zero in all cases); note that scale consistency is a separate question—see Appendix A. “Reversal range” is the range of honest-version spread values for which the Cramér distance reverses the classical CRPS ranking. “BN concordance” indicates whether the Bessac–Naveau conditional-expectation correction (Bessac and Naveau, 2021)—an independent method for the same problem, which computes the expected score over the observation-error distribution—agrees with the Cramér distance on which version wins; high concordance validates the Cramér-distance result.

In all three cases, the Cramér distance has zero location bias at the demonstrated parameters,

produces a ranking reversal when the tight version’s spread is far below the observational uncertainty, and agrees with the Bessac–Naveau correction on the winner.

B.3 UCDP comparison details

Comparison with bounds-based Constructor 1. I also score with Constructor 1 (log-normal F_{obs} fitted to the UCDP low/best/high bounds; §3). Of the 19,397 cell-months, 4,679 have genuine bounds (low $\neq$ best or high $\neq$ best, with low > 0); the remaining 14,718 have degenerate bounds (low = best = high), for which Constructor 1 reduces to a step function and the Cramér distance reduces to classical CRPS. Among cell-months with genuine bounds, the reversal rate is 9.9%—substantially lower than Constructor 3 (39.0%). This reflects the fact that UCDP’s low/high bounds are coding ranges rather than statistical confidence intervals (Vesco et al., 2026): the bounds are narrow relative to the true observational uncertainty, producing an F_{obs} close to the degenerate step function. The bounds-based Constructor 1 therefore understates true observational uncertainty, and the 9.9% reversal rate should be read as a lower bound.

Comparison with the RVI Gumbel mixture. I apply Constructor 2 (the RVI Gumbel mixture defined in §3) and repeat the comparison. Under the RVI constructor, the ranking reversal is substantially wider than under the parametric Constructor 3: the honest version wins in 80.8% of cell-months (versus 39.0% for Constructor 3). This is expected: the RVI model captures asymmetric underreporting—particularly severe for low-fatality events—that the symmetric log-normal parametric constructor cannot represent. The Bessac–Naveau correction confirms the pattern, with 79.6% reversal under RVI draws.

A structural limitation applies, however: the Vesco et al. (2026) regression coefficients predict uncertainty for *individual coded events*, whereas the comparison above aggregates events to cell-months by summing best estimates before applying the RVI constructor. This is a unit-of-analysis mismatch—the model’s vague-number dummies are designed for event-level reported values, not for sums across multiple events. For a cell-month containing many small events, the aggregated sum suppresses the vague-number signal and may understate the Gumbel component’s scale. The direction of this effect is conservative: event-level application would produce *wider* F_{obs} distributions (each small event carries high relative uncertainty), yielding a reversal rate at least as large as the 80.8% reported here. The bounds-based aggregation (Constructor 1, summing low/best/high) has a parallel limitation: summing bounds assumes independent per-event coding errors, which is violated when events in the same cell-month share media sources or coders. A proper event-level analysis that combines event-level F_{obs} distributions into cell-month aggregates is non-trivial and is left to future work; the aggregated results here should be read as illustrative of the direction and approximate magnitude of the effect.

Per-violence-type analysis. Table 5 breaks the comparison down by UCDP violence type. State-based violence ($n = 10,760$) shows the lowest RVI reversal rate (79.3%), consistent with state-based events being better documented and having higher fatality counts on average. Non-state violence ($n = 4,363$) shows the highest reversal rate (90.4%) and is the only type where the RVI Cramér distance selects the honest version *on average*—non-state events tend to have

smaller counts where underreporting is most severe. One-sided violence ($n = 4,003$) falls in between at 84.6%. A caveat: the per-type differences confound documentation quality with event-size distribution. Non-state events are both less well documented *and* smaller on average; the higher reversal rate could reflect either factor or both.

Violence type	n	Classical	Parametric		RVI Gumbel	
		tight %	reversal %	BN agree %	reversal %	BN agree %
State-based	10,760	100%	34.8%	93.3%	79.3%	78.2%
Non-state	4,363	100%	62.9%	94.1%	90.4%	90.2%
One-sided	4,003	100%	43.7%	94.3%	84.6%	84.1%
All	19,397	100%	39.0%	98.1% [†]	80.8%	98.5% [†]

Table 5: Ranking reversal rates by UCDP violence type. “Classical tight %” is the fraction of cell-months where classical CRPS selects the tight version. “Reversal %” is the fraction where the Cramér-distance reformulation reverses the ranking. “BN agree” is the Bessac–Naveau concordance—the fraction of cell-months where the Bessac–Naveau conditional-expectation correction agrees with the Cramér distance on which version wins—computed separately under each F_{obs} constructor (parametric and RVI). The parametric constructor uses $\sigma_{\text{rel}} = 0.4$; the RVI Gumbel uses the regression coefficients of Vesco et al. (2026). Per-type counts do not sum to the pooled total because the historical-statistics join (cells with ≥ 3 active months) applies per type; 271 cell-months qualify in the pooled analysis but fall below the per-type activity threshold. [†]Pooled BN concordance uses $M = 2,000$ Monte Carlo draws; per-type rows use $M = 200$.

The per-type pattern is informative: the Cramér-distance reformulation is most consequential for the violence types where observational uncertainty is greatest, and least consequential where data quality is highest.

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